# CMP 438 Project - Report

**Project Title:** Bluetooth-Controlled Car
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**Course:** CMP 438 - Communicating Robots
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# 1. Introduction

This project focuses on the design, construction, and evaluation of a fully operational Bluetooth-controlled robotic car. The system demonstrates wireless communication using an HC-05 Bluetooth module, embedded programming using Arduino C++, and coordinated motion control through the L298N motor driver. The robotic platform responds to directional commands sent from an Android smartphone running the Arduino Car Connect App, which allows the user to control movement, rotation, and LED lighting through a simple interface.

The robotic vehicle integrates hardware, software, and communication concepts taught in the CMP 438 Communicating Robots course. The final design highlights concepts such as sensorless robot navigation, Bluetooth serial communication, and electronic circuit integration. The overall purpose of this project is to construct a functioning mobile robot capable of receiving wireless input and executing physical actions through programmed logic.

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# 2. Objectives

The primary objectives of the Bluetooth-controlled robotic car project were:

1. **Wireless Control:**
   Implement a reliable Bluetooth communication system enabling remote robot control from an Android smartphone.
2. **Motor Coordination:**
   Use the L298N motor driver to control four DC motors and support forward, backward, turning, pivoting, and stopping behaviors.
3. **Embedded Programming:**
   Develop and test Arduino C++ code to interpret Bluetooth commands, execute actions, print diagnostic messages, and manage motor output.
4. **Physical Assembly and Wiring:**
   Construct a stable robotic chassis powered by a 7.4V Li-Ion battery. Integrate all wiring and ensure correct power distribution among components.

5. **Functional Demonstration:**
   Record a full demonstration video to confirm the robot moves reliably under wireless control.

View the complete Arduino code here:
https://github.com/Chaotic-Legend/CMP-438-Codes/blob/main/Project%3A%20Bluetooth-Controlled%20Car/CMP%20438%20Project%20-%20Bluetooth-Controlled%20Car.ino

View the working TinkerCAD circuit here:
https://www.tinkercad.com/things/igILUXtGBrG-cmp-438-project-keyboard-controlled-car?sharecode=ouXh349QThBm-k8ye4wscJ36WKpGQb2r2AGfotNXZYA

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# 3. Components Used

- Inland Uno R3 Arduino-Compatible MainBoard
- 4 × TT DC 3–6V Dual-Shaft Gearbox Motors (200 RPM, 1:48 Ratio)
- 4 × Tire Wheels
- L298N Dual H-Bridge Motor Driver
- HC-05 Bluetooth Module
- Blue LED
- 7.4V Li-Ion Battery Pack
- Plastic Chassis
- Jumper Wires (Male–Male and Female–Male)

These components were chosen based on their compatibility, availability, and suitability for small robotic platforms.

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# 4. System Design Overview

The robotic system consists of three major subsystems:

1. **Control Subsystem** – Arduino UNO and HC-05 module
2. **Actuation Subsystem** – Four DC motors controlled via L298N
3. **Power Subsystem** – 7.4V Li-Ion battery powering motors and logic lines

Bluetooth commands from the smartphone are sent through the HC-05 module to the Arduino UNO, which interprets each character command and activates the motor driver pins accordingly. An LED connected to pin 13 provides visual feedback for movement and lighting commands.

## 5. Wiring Documentation

### HC-05 Bluetooth Module

| HC-05 Pin | Arduino Pin | Description |
|---|---|---|
| VCC | 5V | Provides regulated power to the Bluetooth module. |
| GND | GND | Establish common ground across the system. |
| TXD | RX (0) | Sends Bluetooth data to the Arduino. |
| RXD | TX (1) | Receives data from the Arduino if needed. |

### L298N Motor Driver

| L298N Pin | Arduino Pin | Description |
|---|---|---|
| IN1 | 8 | Controls the forward motion of the right motor. |
| IN2 | 9 | Controls the backward motion of the right motor. |
| IN3 | 10 | Controls the forward motion of the left motor. |
| IN4 | 11 | Controls the backward motion of the left motor. |
| GND | GND | Common ground connection. |

## 6. Software Design

The Arduino program initializes all motor and LED pins, establishes serial communication at 9600 baud, and enters a loop waiting for Bluetooth input. When a command character is received, it matches a corresponding movement or action.

| Command | Behavior |
|---|---|
| F | Move Forward |
| B | Move Backward |
| L | Turn Left |

| | |
|---|---|
| R | Turn Right |
| A | Pivot Left |
| C | Pivot Right |
| S | Stop |
| O | LED On |
| s | LED Off |

Each action triggers a motor-control function and prints a status message to the Serial Monitor. The final code includes clear comments explaining every line for instructional purposes.

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# 7. Simulation Overview

We developed a TinkerCAD simulation to demonstrate basic functionality using keyboard input instead of Bluetooth. The simulation successfully demonstrates direction changes and stopping behaviors using the same motor control logic. Due to the limitations of TinkerCAD, the complete physical system could not be accurately replicated, particularly because of the lack of HC-05 support and a limited number of motors.

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# 8. Responsibilities

- **Isaac D. Hoyos:**
  Developed all Arduino code, created the TinkerCAD simulation, wrote the documentation, designed the wiring diagrams, prepared the GitHub repository, and authored the complete written report.
- **Roberto Morales:**
  Assembled the physical robot, wired all components, installed the motors, tested the Arduino code on hardware, and recorded the demonstration video.

The final system resulted from coordinated collaboration between both team members.

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# 9. Results and Performance Evaluation

After assembling the robotic vehicle and uploading the final Arduino code, the robot performed reliably and consistently under Bluetooth control. The Arduino Car Connect App successfully transmitted directional commands to the HC-05 module, which the Arduino processed without delay.

- **Forward and Backward:**
  Pressing the forward button or the "F" command caused smooth forward motion. Pressing the backward button or "B" reversed the motors correctly.
- **Turning:**
  The robot responded accurately to left and right turn commands using directional buttons or "L" and "R."
- **Pivoting:**
  Dedicated buttons enabled clockwise and counterclockwise rotation, allowing the car to pivot in place.
- **LED Control:**
  A blue LED can be turned on or off through the app using the lighting command, demonstrating an additional functional channel through Bluetooth.
- **Stopping Behavior:**
  The robot halted instantly when given the stop command, and it also came to a safe stop when no directional command was sent through the smartphone interface.

Overall, the robot demonstrated smooth, predictable, and intuitive control, confirming the success of the Bluetooth communication, motor driver circuitry, wiring layout, and embedded programming. Bluetooth remained stable at several meters of distance, and the motors operated without stalling.

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# 10. Conclusion

This Bluetooth-controlled robotic car project successfully met all objectives established at the beginning of the semester. The system effectively integrates embedded programming, motor driver control, wireless communication, and practical robotics assembly into a fully functioning mobile platform. The robot demonstrated reliable performance, clear responsiveness to commands, and robust hardware integration.

The project provided valuable hands-on experience in designing, assembling, coding, and evaluating a robotic system. It also reinforced essential skills in troubleshooting, teamwork, and documentation. The final result is a well-designed robotic car that demonstrates the main concepts taught in CMP 438: Communicating Robots.

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